benchdamic: benchmarking of differential abundance methods on microbiome data

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THE ISSUE

Differential abundance (DA) analysis is one of the primary approaches to identify differences in the microbial community composition between samples and to understand the structures of the microbial communities and the associations between the microbial compositions and the environment.

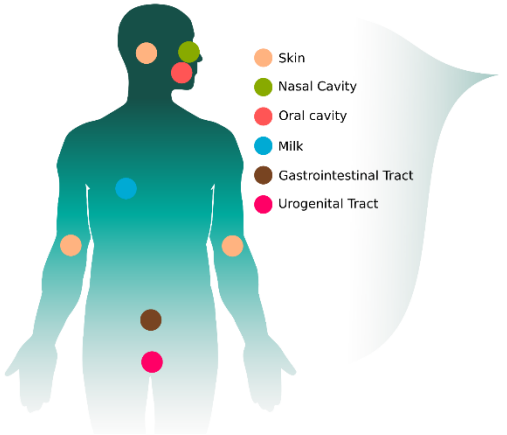
The correct identification of differentially abundant microbial taxa between experimental conditions is a methodological and computational challenge and, as is often the case in high-throughput biology, the results are data-dependent.

- Too many zeros
- No FDR control
- Wrong model assumptions
- Data specific results

THE IDEA

On several curated datasets, we compare methods developed for single-cell and bulk RNA-seq, and specifically for metagenomics.

Conclusions are based on several performance metrics: the ability for some statistical distributions to successfully fit the sparsity level and the counts, the ability of the methods to control the type I error, the concordance among methods' results, and, if a priori knowledge is available, the enrichment test.



Number of datasets

16S x 18

WMS x 82

2 Repositories

HMP16SData

CuratedMetagenomicData

41 Projects

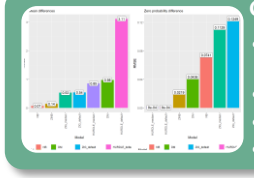
CONCLUSIONS

- Calgaro et al. 2020*
- Exploratory data analysis
- Dataset specific choices

THE APPLICATION

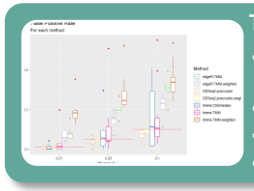
Benchdamic, a R package available on Bioconductor, makes more accessible the benchmark framework described by our previous work (Calgaro *et al.* 2020*) providing a tool to guide researchers towards an informed decision when evaluating the best suitable methods to use for their data.

The package allows the users to test a variety of commonly used methods for differential abundance analysis, including also custom methods, on their own datasets. Benchdamic represent therefore a flexible and customizable benchmarking platform that can be used as a framework to evaluate new differential abundance detection methods.



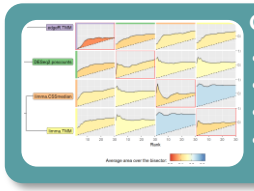
Goodness of Fit

- **Question:** Which are the parametric distributions that are able to fit both the proportion of zeros and the counts in your data?
- **Input:** phyloseq object or count matrix
- **Working function:** `FitModels`
- **Output:** `plotRMSE`, `plotMD`



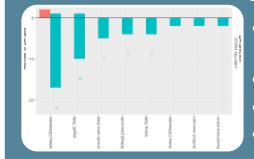
Type I Error Control

- **Question:** Which are the DA methods that are able to control the number of false positives in your data?
- **Input:** phyloseq object
- **Working functions:** `createMocks`, `runMocks`, `createTIEC`
- **Output:** `plotFPR`, `plotKS`, `plotQQ`



Concordance

- **Question:** If multiple DA methods are run on the same data, would they be concordant?
- **Input:** phyloseq object
- **Working functions:** `createSplits`, `runSplits`, `createConcordance`
- **Output:** `plotConcordance`



Enrichment

- **Question:** If some prior knowledge about your experiment is available, would the findings be coherent with that knowledge?
- **Input:** phyloseq object, some a priori knowledge
- **Working functions:** `runDA`, `createEnrichment`, `createPositives`
- **Output:** `plotEnrichment`, `plotContingency`, `plotMutualFindings`, `plotPositives`

* Calgaro M. *et al.* Assessment of statistical methods from single cell, bulk RNA-seq, and metagenomics applied to microbiome data. *Genome Biol* **21,** 191 (2020). https://doi.org/10.1186/s13059-020-02104-1